



Checking understanding of sequences

Task A: Sequences with a constant additive pattern

- 1) These pupils are making number patterns from a rule



Aisha



Andeep



Izzy



Jacob



Jun



Lucas



Laura



Sam



Alex



Sofia

Aisha says any number, Andeep adds a number to it. Each pupil then adds the same number Andeep added. For each question, justify your answer.

- a) If Aisha says 6 and Andeep says 10, what will Izzy say?
- b) If Aisha says 6 and Andeep says 10, what will Sofia say?
- c) After a different starting number Sam says 17 and Alex says 19. What did Aisha say?
- d) If Aisha says 4 to start and Jun says 40 What will Lucas say?
- e) Aisha starts with 10, only Lucas' number is another integer, what could Andeep have said?

Task B: Sequences with a constant multiplicative pattern

1) Sort these sequences into the correct column

Sequences with a constant additive relationship	Sequences with a constant multiplicative relationship

5, 10, 15, 20, ...

5.7, 8.1, 10.5, 12.9, ...

3, -6, 12, -24 ...

1, 2, 4, 8, ...

48, 24, 12, 6, ...

6, 48, 90, 132, ...

3, 30, 300, 3000, ...

1.5, 0.5, -0.5, -1.5, ...

0.5, 1.5, 4.5, 13.5, ...

2) Find the missing numbers in these sequences with constant multiplicative patterns.

a) 6, 12, 24, ____, ...

e) ____, 21, 63, 189, ...

b) 5, 30, ____, 1080, ...

f) 1, 0.2, 0.04, ____, ...

c) 160, 80, 40, ____, ...

g) 0.4, 8, 160, ____, 64 000, ...

d) 0.3, -3, 30, ____, ...

h) ____, 6, 8, $\frac{32}{3}$, ...